

Section B: Endpoint API plan

HTT P Met hod	Route	Descrip tion	Role Requ ired	Request Body	Expected Response
POST	/api/auth/register	Register a new user account in the system. Creates either an Organizer or Participant profile based on role.	None (public)	<pre>{ "email": "string", "password": "string", "firstName": "string", "lastName": "string", "phoneNumber": "string", "role": "Organizer/Participant", "dateOfBirth": "date", "gender": "M/F/O", "idNumber": "string", "emergencyContactName": "string", "emergencyContactNumber": "string", "medicalConditions": "string (optional)", "clubAffiliation": "string (optional)" }</pre>	201 Created - User object with JWT token; 400 Bad Request - Validation errors; 409 Conflict - Email already exists
POST	/api/auth/login	Authenticate user credentials and return a JWT access token for later API calls.	None (public)	<pre>{ "email": "string", "password": "string" }</pre>	200 OK - {"token": "jwt_token", "user": {user details}}; 401 Unauthorized - Invalid email or password

POST	/api/auth/logout	Invalidate the current user session by blacklisting the JWT token.	Any (logged in)	None	200 OK - {"message": "Logged out successfully"}; 401 Unauthorized - No active session
GET	/api/auth/me	Retrieve the profile information of the currently authenticated user.	Any (logged in)	None	200 OK - User profile with role-specific details (organizer or participant); 401 Not authenticated

GET	/api/users/profile	Get the complete profile of the authenticated user, including role-specific data.	Any (logged in)	None	200 OK - Complete user profile with role-specific details; 404 Not Found - User not found
PUT	/api/users/profile	Update the authenticated user's profile information. Fields can be updated individually.	Any (logged in)	{ "firstName": "string", "lastName": "string", "phoneNumber": "string", "email": "string" }	200 OK - Updated user profile; 400 Bad Request - Validation errors; 404 Not Found - User not found

PUT	/api/users/password	Change the authenticated user's password. Requires current password verification.	Any (logged in)	{ "currentPassword": "string", "newPassword": "string" }	200 OK - {"message": "Password updated successfully"}; 400 Bad Request - Invalid current password or weak new password
GET	/api/users/role-specific	Get role-specific profile details (Organiser details or Participant details) based on user roles.	Any (logged in)	None	200 OK - Role-specific details object; 400 Bad Request - Role type mismatch or invalid role

GET	/api/events	Retrieve a list of all events with optional filtering by city, province, event type, date range, and status. Supports pagination.	None (public)	<i>Query Params:</i> city, province, eventType, dateFrom, dateTo, status, page, limit	200 OK - {"events": [{event objects}], "total": count, "page": page, "limit": limit}
GET	/api/events/{id}	Get detailed information about a specific event including its categories and organizer details.	None (public)	None	200 OK - Event details with categories and enrolment counts; 404 Not Found - Event not found

POST	/api/events	Create a new event. The authenticated user must have the Organiser role.	Organiser	{ "title": "string", "description": "string", "eventDate": "date", "eventTime": "time", "venue": "string", "city": "string", "province": "string", "eventType": "Running/Walking/Cycling/Triathlon/Multi-Sport", "maxParticipants": "int (optional)", "registrationOpenDate": "datetime (optional)", "registrationCloseDate": "datetime (optional)", "metadata": "object (optional)" }	201 Created - New event object; 400 Bad Request - Validation errors; 403 Forbidden - User is not an Organiser
PUT	/api/events/{id}	Update an existing event. Only the event organiser can change their own events.	Organiser	{ "title": "string", "description": "string", "eventDate": "date", "eventTime": "time", "venue": "string", "city": "string", "province": "string", "eventType": "string", "status": "Planning/Open/Closed/In Progress/Completed/Cancelled", "maxParticipants": "int", "registrationOpenDate": "datetime", "registrationCloseDate": "datetime" }	200 OK - Updated event object; 400 Bad Request - Validation errors; 403 Forbidden - Not the event organiser; 404 Not Found - Event not found

DELETE	/api/events/{id}	Delete an event and all its associated categories, enrolments, and results.	Organiser	None	204 No Content - Event dropped successfully; 403 Forbidden - Not the event organiser; 404 Not Found - Event not found
GET	/api/events/{id}/categories	Retrieve all categories associated with a specific event including enrolment counts.	None (public)	None	200 OK - List of categories with enrolment statistics; 404 Not Found - Event not found

GET	/api/events/{id}/stats	Get comprehensive statistics for an event including total enrolments, category breakdowns, and participant demographics.	None (public)	None	200 OK - {"totalEnrolments": int, "categories": [{category stats}], "demographics": {gender, age groups}}; 404 Not Found - Event not found
GET	/api/categories	Retrieve all categories with optional filtering by event ID, category type, or distance range.	None (public)	<i>Query Params:</i> eventId, categoryType, distanceMin, distanceMax	200 OK - List of categories with event details and enrolment counts

GET	/api/categories/{id}	Get detailed information about a specific category including enrolment count and event details.	None (public)	None	200 OK - Category details with enrolment count; 404 Not Found - Category not found
POST	/api/events/{eventId}/categories	Add a new category to an existing event. Only the event organiser can add categories.	Organiser	{ "categoryName": "string", "description": "string", "distance": "decimal", "distanceUnit": "km/miles", "ageGroupMin": "int (optional)", "ageGroupMax": "int (optional)", "genderRestriction": "M/F/O (optional)", "entryFee": "decimal", "lateEntryFee": "decimal (optional)", "maxParticipants": "int (optional)", "startTime": "time (optional)", "cutoffTime": "time (optional)" }	201 Created - New category object; 400 Bad Request - Validation errors; 403 Forbidden - Not event organiser; 404 Not Found - Event not found

PUT	/api/categories/{id}	Update an existing category . Only the event organiser can change the categories.	Organiser	{ "categoryName": "string", "description": "string", "distance": "decimal", "distanceUnit": "km/miles", "ageGroupMin": "int", "ageGroupMax": "int", "genderRestriction": "M/F/O", "entryFee": "decimal", "lateEntryFee": "decimal", "maxParticipants": "int", "startTime": "time", "cutOffTime": "time" }	200 OK - Updated category object; 400 Bad Request - Validation errors; 403 Forbidden - Not event organiser; 404 Not Found - Category not found
DELETE	/api/categories/{id}	Delete a category . Only the event organiser can delete categories.	Organiser	None	204 No Content - Category deleted; 403 Forbidden - Not event organiser; 404 Not Found - Category not found

GET	/api/enrolments	Retrieve all enrolments with optional filtering by participant, event, status, or date range.	Participant (own), Organiser (all)	<i>Query Params:</i> participantId, eventId, status, dateFrom, dateTo	200 OK - List of enrolments; 403 Forbidden - Access denied (trying to access other's enrolments)
GET	/api/enrolments/{id}	Get detailed information about specific enrolments including participants and category details.	Participant (own), Organiser (all)	None	200 OK - Enrolment details with participant and category info; 403 Forbidden - Access denied; 404 Not Found - Enrolment not found

POST	/api/events/{eventId}/enrol	Enrol a participant in a specific event category. Participants must be authenticated.	Participant	<pre>{ "categoryId": "int", "emergencyContactName": "string (optional)", "emergencyContactNumber": "string (optional)", "medicalConditions": "string (optional)", "notes": "string (optional)" }</pre>	
-------------	-----------------------------	--	-------------	--	--